Exam December 2021 Question paper. Part 1.

Physics for Engineers (University of Pretoria)

FSK 176

Examination - Part 2 of 2

Department of Physics, University of Pretoria

Date: 2021-12-06 **Time: 2 hours** **Total: 58**

Total time for the examination (Part 1 and Part 2): 3 hours

Examiner: Prof. JM Nel **Moderator: Prof. WE Meyer**

Please read the following carefully.

1. Answer all the questions on lined or blank paper.
2. Once you are done, scan your answers into a pdf document and submit your answers on Gradescope.
3. This is an open book test - you are allowed to use your textbook when answering this test. You may not search the internet for solutions to the problems.
4. Where applicable you need to draw the necessary labelled diagrams to assist with answering the question - such as free body diagrams.
5. Your answer needs to be written in a systematic and logical fashion. Tell us what you are doing. You must show that you understand and can use the relevant physics. **Only substitute numerical values in the final step of calculations! A correct answer does not imply that you will receive full marks**.
6. **Units must be used when numbers are substituted else marks will be subtracted.**
7. All exam regulations of the University of Pretoria apply, and any contravention of it will be treated in a serious manner.

Declaration:

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment
In proceeding with this test, you are accepting the contents of the above declaration.

Constants and formulas
$G = 6.674 \times 10^{-11}$ N·m^2/kg^2; $e = 1.602 \times 10^{-19}$ C; $k = 1.381 \times 10^{-23}$ J/K; $m_e = 9.109 \times 10^{-31}$ kg; $m_p = 1.673 \times 10^{-27}$ kg; $R = 8.314$ J/(mol·K); $N_A = 6.02 \times 10^{23}$ /mol; 1 atm = 101.3 kPa; 1 u = $1.6605677 \times 10^{-27}$ kg; $c = 3 \times 10^8$ m/s; $V_{\text{sphere}} = \frac{4}{3}\pi r^3$; $F_{gravity} = \frac{Gm_1m_2}{r^2}$; 1 litre = 1000 cm^3; $M_{\text{Earth}} = 5.97 \times 10^{24}$ kg, $\rho(\text{water}) = 1$ g/cm^3; $r_{\text{Earth}} = 6.37 \times 10^6$ m

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1 Question 1 [8]

The position of a 2 kg object is given as $x(t) = \beta t^2 + 5$, where x is in meters and t is in seconds.

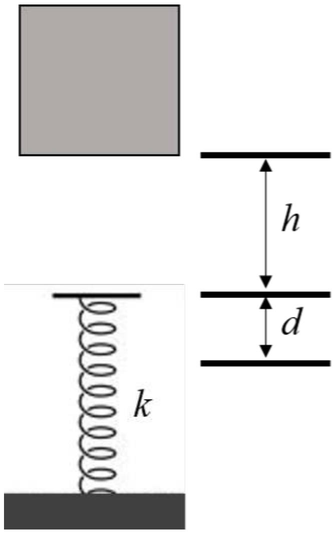
(a) Determine the force F responsible for this motion. (3)

(b) If the force only results in a change in the kinetic energy of the object of 200 J between $t_1 = 0$ s and $t_2 = 5$ s, determine the value of β. (3)

(c) Without using kinematics (equations of motion), determine the displacement of the object during these 5 seconds referred to in part (b). (2)

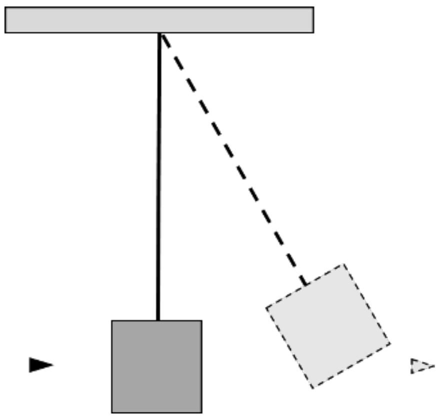
2 Question 2 [6]

A spring, with spring constant $k = 1960$ N, is mounted vertically on a table as shown in the figure. A block of mass $m = 2.0$ kg is dropped from a height h and falls on the spring compressing it by a distance $d = 11$ cm. From what height, h, above the spring, was the block released from rest?



3 Question 3 [12]

A bullet with a speed of 300 m/s passes right through a block with a mass of 400 g. The block is suspended on a long cord. The impulse gives the block a speed of 1.5 m/s.



(a) If the bullet has a mass of 3 g, determine the speed of the bullet after it has passed through the block. (3)

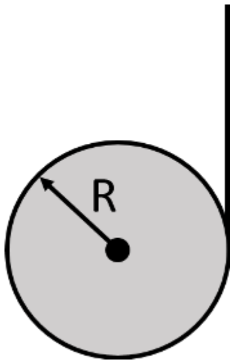
(b) By what vertical distance will the center of mass of the block rise as the block swings upward after the bullet passes through it? (4)

(c) Determine the work done by the bullet as it passes through the block. (2)

(d) How much of the system's mechanical energy (bullet + block) is converted to heat? (3)

Question 4 [9]

A string is wound around a homogeneous, horizontal cylinder of mass M and radius R. The cylinder is then held above the ground and released from rest. As the string unwinds and the cylinder spins, the end of the string is continually pulled vertically upwards with a force that is just sufficient to keep the cylinder from descending to the ground. The figure shows the situation viewed from the side.



(a) Identify the forces acting on the system (string and cylinder) and determine an expression for the tension in the vertical portion of the string. (3)

(b) Determine the angular acceleration of the cylinder. (4)

(c) What is the upward acceleration of any point along the vertical portion of the string? (2)

Question 5 [5]

The diaphragm of a loudspeaker oscillates in order to produce sound. The amplitude of such an oscillation has been found to be 5.00 μm.

(a) At what frequency will the magnitude of the diaphragm's acceleration be equal to g. (3)

(b) What will the effect be on the acceleration if the frequency is doubled. Explain your answer. (2)

Question 6 [9]

Two sinusoidal waves in a string are defined by the wave functions

$$y_1 = 2.00 \sin(20.0x - 32.0t)$$

$$y_1 = 2.00 \sin(25.0x - 40.0t)$$

where x, y_1 and y_2 are in centimeters, t is in seconds and the angles are measured in radians.

(a) What is the phase difference between these two waves at the point $x = 5.00$ cm at $t = 2.00$ s. Give your final answer in degrees between 0^{O} and 360^{O}. (4)

(b) What is the positive x value closest to the origin for which the two phases differ by $\pm\pi$ at $t = 2.00$ s resulting in a zero displacement? Show all your workings and explain your reasoning. (5)

Question 7 [9]

A lead bullet is fired into a wall and is embedded. A myth claims that a bullet can be completely melted due to friction. A friend of yours doubts that this is possible. Using your physics knowledge you set out to test the validity of the myth. The first assumption that you make is that **ALL** kinetic energy of the bullet is converted into heat due to friction and the second assumption is that it was a pleasant day with an ambient temperature of 25.0^{O}C. Determine the minimum speed of the lead bullet that will be required to achieve the outcome of the myth. Is this myth of the melting lead bullet plausible if you consider that the average bullet can reach speeds of 700 m/s?
The latent heat of fusion for lead is 2.32×10^4 J/kg, the specific heat of lead is 128 J/(kg$\cdot^{\mathrm{O}}$C) and the melting point of lead is 327^{O}C.

The end!